

Forward Guidance and the Policy Rate Path: Transparency, Type Uncertainty, and the Exchange Rate^{*}

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Abstract

We develop a two-period small open economy New Keynesian model to compare explicit forward guidance with opacity when the central bank's inflation aversion is private information. Under opacity, uncertainty about the central bank's reaction function generates exchange rate misalignment and suboptimal period-1 allocations. Forward guidance—modelled as a non-binding rate forecast—resolves this uncertainty, improving welfare through two channels: an exchange rate channel specific to the open-economy setting, and an expectation coordination channel. The welfare gain is strictly positive when cost-push shocks are present and type uncertainty is genuine, and is non-monotone in the degree of openness.

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1 Introduction

Should central banks publish their interest rate forecasts? The question sits at the heart of modern monetary policy communication. Since the Reserve Bank of New Zealand first released an explicit rate path in 1997, a growing number of central banks have adopted the practice of *forward guidance*—communicating, in varying degrees of detail, the expected future path of the policy rate. Yet a parallel tradition of institutional opacity persists, and the welfare consequences of these two approaches remain contested.

This paper develops a two-period small open economy New Keynesian model to compare two communication regimes: *explicit forward guidance*, in which the central bank announces a forecast of the period-2 interest rate; and *opacity*, in which it remains silent about future intentions. A key feature of the model is that the central bank’s inflation aversion—its “type”—is private information. Private agents observe current macroeconomic conditions but do not know whether they face a hawkish or a dovish policymaker. This uncertainty about the reaction function generates misaligned expectations about future interest rates and, through the uncovered interest parity (UIP) condition, misaligned exchange rate expectations in the current period.

Forward guidance resolves this uncertainty by credibly revealing the central bank’s type. We show that this has two distinct welfare-improving channels: (i) it reduces the variance of private-sector forecast errors about the period-2 rate, lowering expected losses in that period; and (ii) it corrects exchange rate misalignment in the current period, improving period-1 allocations. The exchange rate channel is specific to the open-economy setting and amplifies the value of transparency relative to what a closed-economy analysis would imply.

[The introduction will be extended once results are fully derived.]

Related literature. [Placeholder for literature review.]

Outline. Section 2 presents the model environment, defines the two polar communication regimes, and characterizes equilibrium in each period. Section 3 compares welfare under forward guidance and opacity. Section 4 extends the analysis to two intermediate regimes—the constant-rate conditional forecasts of the SNB and the market-rate conditional forecasts of the BOE—and characterizes scenario-based communication. Section 5 concludes.

2 Model

2.1 Environment

Time is discrete with two periods, $t \in \{1, 2\}$. The economy is a small open economy populated by a representative household-firm unit. The central bank sets the nominal interest rate i_t each period to minimise a quadratic loss function. We use lower-case variables to denote log-deviations from steady state; all steady-state values are normalised to zero.

IS curve. Aggregate demand in each period follows the standard open-economy Euler equation:

$$y_t = \mathbb{E}_t[y_{t+1}] - \sigma(i_t - \mathbb{E}_t[\pi_{t+1}]) + \alpha q_t + \varepsilon_{d,t}, \quad (1)$$

where y_t is the output gap, π_t is CPI inflation, i_t is the nominal policy rate, q_t is the (log) real exchange rate (an increase is a depreciation), and $\varepsilon_{d,t}$ is a demand disturbance. The parameter $\sigma > 0$ is the intertemporal elasticity of substitution and $\alpha > 0$ captures the expenditure-switching effect of the real exchange rate on net exports.

New Keynesian Phillips curve. Inflation dynamics are governed by

$$\pi_t = \beta \mathbb{E}_t[\pi_{t+1}] + \kappa y_t + \varepsilon_{c,t}, \quad (2)$$

where $\beta \in (0, 1)$ is the household discount factor, $\kappa > 0$ is the slope of the Phillips curve, and $\varepsilon_{c,t}$ is a cost-push shock.

Uncovered interest parity. The real exchange rate satisfies the forward-looking UIP condition:

$$q_t = \mathbb{E}_t[q_{t+1}] - (i_t - i^*), \quad (3)$$

where $i^* \geq 0$ is the (exogenous, constant) foreign interest rate. Because the model has two periods, the exchange rate today embeds the entire expected path of future interest rate differentials. Iterating (3) forward and imposing the terminal condition $\mathbb{E}_2[q_3] = 0$ gives:

$$q_1 = -(i_1 - i^*) - \mathbb{E}_1[i_2 - i^*]. \quad (4)$$

Equation (4) is central to the paper's mechanism: the current exchange rate is pinned by the entire expected path of the policy rate. Any uncertainty about i_2 therefore transmits immediately into q_1 .

Terminal conditions. Period 2 is the final period. We impose:

$$\mathbb{E}_2[y_3] = \mathbb{E}_2[\pi_3] = \mathbb{E}_2[q_3] = 0. \quad (5)$$

Substituting (5) into (1)–(2) yields the period-2 block:

$$y_2 = -\sigma i_2 + \alpha q_2 + \varepsilon_{d,2}, \quad (6)$$

$$\pi_2 = \kappa y_2 + \varepsilon_{c,2}. \quad (7)$$

From (3) and the terminal condition:

$$q_2 = -(i_2 - i^*). \quad (8)$$

Substituting (8) into (6) and defining the *effective interest rate sensitivity*

$$\tilde{\sigma} \equiv \sigma + \alpha > \sigma, \quad (9)$$

the period-2 output gap (after normalising $\alpha i^* = 0$) becomes:

$$y_2 = -\tilde{\sigma} i_2 + \varepsilon_{d,2}. \quad (10)$$

The parameter $\tilde{\sigma}$ exceeds the closed-economy elasticity σ because a rate increase both suppresses domestic spending *and* appreciates the exchange rate, reducing net exports.

2.2 Central Bank Preferences and Type

The central bank minimises the standard quadratic loss function:

$$\mathcal{L} = \sum_{t=1}^2 \beta^{t-1} \left[y_t^2 + \lambda \pi_t^2 \right], \quad (11)$$

where $\lambda > 0$ is the relative weight on inflation stabilisation. A higher λ represents a more *hawkish* central bank; a lower λ a more *dovish* one.

Private information. The inflation weight λ is the central bank's *private information*.

The private sector holds a prior over two possible types:

$$\lambda \in \{\lambda_L, \lambda_H\}, \quad \lambda_L < \lambda_H, \quad (12)$$

with prior probabilities

$$\Pr(\lambda = \lambda_H) = p \in (0, 1), \quad \Pr(\lambda = \lambda_L) = 1 - p. \quad (13)$$

Private agents know the distribution (13) and the model equations but cannot directly observe λ .

Remark. The assumption that λ is private information is motivated by empirical evidence that central bank preferences are not perfectly observable to the public: markets spend considerable resources inferring the degree of inflation aversion from policy decisions, communications, and appointments. This uncertainty about the reaction function—rather than uncertainty about economic fundamentals—is the source of welfare losses in the opacity regime.

2.3 Timing and Communication Regimes

The sequence of events is as follows:

1. **Period 1 (announcement stage).** The cost-push shock ε_c is publicly observed.¹

The central bank sets i_1 . Under the two regimes:

- *Forward guidance (FG):* The central bank announces $\hat{i}_2$, a forecast of its intended period-2 rate. The announcement is truthful and believed in full by private agents.
- *Opacity (OP):* The central bank makes no announcement. Private agents form $\mathbb{E}_1^{\text{OP}}[i_2]$ using Bayes' rule over the type distribution.

2. **Period 2 (implementation stage).** The demand shock ε_d is realised and observed by all. The central bank sets i_2 by minimising its period-2 loss, taking the realised shocks and its own type as given. There is *no commitment*: the bank is free to deviate from any period-1 announcement.

Under *forward guidance*, the announcement $\hat{i}_2$ is the central bank's rational forecast of

¹This captures the realistic scenario in which the central bank and private sector alike observe current inflationary pressure—for example, an energy price spike—before the bank announces its policy intentions.

its own period-2 decision, formed in period 1 before ε_d is observed:

$$\hat{i}_2(\lambda) \equiv \mathbb{E}_1[i_2^*(\lambda, \varepsilon_d, \varepsilon_c)] = \phi_c(\lambda) \varepsilon_c, \quad (14)$$

where the second equality follows from $\mathbb{E}_1[\varepsilon_d] = 0$ and the period-2 optimal policy derived below (Proposition 1). The announcement is a forecast, not a promise: the bank will re-optimize in period 2 upon observing ε_d .

2.4 Period-2 Equilibrium

Proposition 1 (Period-2 Optimal Policy). *In period 2, the central bank of type λ minimises $y_2^2 + \lambda\pi_2^2$ subject to (10) and (7). The optimal policy rate is:*

$$i_2^*(\lambda) = \phi_d \varepsilon_{d,2} + \phi_c(\lambda) \varepsilon_c, \quad (15)$$

where

$$\phi_d \equiv \frac{1}{\tilde{\sigma}}, \quad (16)$$

$$\phi_c(\lambda) \equiv \frac{\lambda\kappa}{\tilde{\sigma}(1 + \lambda\kappa^2)}. \quad (17)$$

Proof. Substituting (10) into (7):

$$\pi_2 = -\kappa\tilde{\sigma} i_2 + \kappa\varepsilon_{d,2} + \varepsilon_c.$$

The first-order condition $\partial(y_2^2 + \lambda\pi_2^2)/\partial i_2 = 0$ gives the optimal trade-off condition:

$$y_2 + \lambda\kappa\pi_2 = 0. \quad (18)$$

Substituting $y_2 = -\tilde{\sigma}i_2 + \varepsilon_d$ and $\pi_2 = -\kappa\tilde{\sigma}i_2 + \kappa\varepsilon_d + \varepsilon_c$ into (18) and solving for i_2 yields (15).

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Interpretation of the response coefficients. Several properties of Proposition 1 deserve emphasis.

- (i) *Demand shocks are accommodated identically by all types.* The coefficient $\phi_d = 1/\tilde{\sigma}$ is independent of λ . Because a demand shock raises output and inflation proportionally, it creates no tension between stabilisation objectives; both types respond by raising the rate to fully close the output gap.
- (ii) *Cost-push shocks reveal the central bank's type.* The coefficient $\phi_c(\lambda)$ is strictly increasing in λ :

$$\frac{\partial \phi_c}{\partial \lambda} = \frac{\kappa}{\tilde{\sigma}(1 + \lambda\kappa^2)^2} > 0. \quad (19)$$

A hawkish central bank (λ_H) raises the rate more aggressively in response to a cost-push shock than a dovish one (λ_L), accepting a larger output contraction to contain inflation. It is therefore the *response to cost-push shocks* that identifies the type.

- (iii) *The exchange rate amplifies the policy transmission.* The effective sensitivity $\tilde{\sigma} = \sigma + \alpha > \sigma$ means that, for a given change in i_2 , output and inflation move more than in a closed economy. This amplification is internalised by the central bank: a smaller rate move is needed to achieve the same stabilisation, so i_2^* is lower in absolute value than its closed-economy counterpart.

Period-2 equilibrium outcomes. Substituting (15) back into (10) and (7):

$$y_2^*(\lambda) = \frac{-\lambda\kappa}{1 + \lambda\kappa^2}(\kappa\varepsilon_{d,2} + \varepsilon_c) + \frac{\varepsilon_{d,2}}{1 + \lambda\kappa^2} \cdot \frac{1}{1}, \quad (20)$$

more compactly:

$$y_2^*(\lambda) = \frac{\varepsilon_{d,2} - \lambda\kappa\varepsilon_c}{1 + \lambda\kappa^2}, \quad (21)$$

$$\pi_2^*(\lambda) = \frac{\kappa\varepsilon_{d,2} + \varepsilon_c}{1 + \lambda\kappa^2}. \quad (22)$$

Note from (22) that the denominator $(1 + \lambda\kappa^2)$ is increasing in λ : a hawkish central bank achieves smaller inflation deviations in response to cost-push shocks, but at the cost of larger output gap deviations (equation (21)).

2.5 Period-1 Equilibrium

We now characterise the period-1 equilibrium under each communication regime. The central bank sets i_1 to minimise total expected loss (11), taking as given the private sector's expectations (which differ across regimes) and the period-2 equilibrium (15).

Expectations under each regime. The period-2 rate is $i_2^*(\lambda) = \phi_d\varepsilon_d + \phi_c(\lambda)\varepsilon_c$. Since $\mathbb{E}_1[\varepsilon_d] = 0$, the critical object is how agents form $\mathbb{E}_1[i_2]$:

- **Forward guidance:** The announcement $\hat{i}_2(\lambda) = \phi_c(\lambda)\varepsilon_c$ is observed and believed.

Thus:

$$\mathbb{E}_1^{\text{FG}}[i_2] = \phi_c(\lambda)\varepsilon_c. \quad (23)$$

- **Opacity:** Agents do not observe λ . Using the prior (13) and the known equilibrium mapping (15):

$$\mathbb{E}_1^{\text{OP}}[i_2] = [p\phi_c(\lambda_H) + (1-p)\phi_c(\lambda_L)]\varepsilon_c \equiv \bar{\phi}_c\varepsilon_c, \quad (24)$$

where $\bar{\phi}_c \equiv p\phi_c(\lambda_H) + (1-p)\phi_c(\lambda_L)$ is the prior-weighted average response coefficient.

Exchange rate in period 1. From (4) and normalising $i^* = 0$:

$$q_1^r = -i_1 - \mathbb{E}_1^r[i_2], \quad r \in \{\text{FG}, \text{OP}\}. \quad (25)$$

The exchange rate gap between regimes is therefore:

$$q_1^{\text{FG}} - q_1^{\text{OP}} = -(\phi_c(\lambda) - \bar{\phi}_c)\varepsilon_c. \quad (26)$$

For a hawkish central bank ($\lambda = \lambda_H$), we have $\phi_c(\lambda_H) > \bar{\phi}_c$, so $q_1^{\text{FG}} < q_1^{\text{OP}}$: the announcement of a hawkish type *appreciates* the exchange rate relative to the opacity benchmark. The opposite holds for a dovish type. Under opacity, the exchange rate is anchored to the prior-weighted average response—a systematic misalignment whenever $\lambda \neq \bar{\lambda}$ (where $\bar{\lambda} = p\lambda_H + (1-p)\lambda_L$).

Period-1 IS curve. We substitute $\mathbb{E}_1[y_2]$, $\mathbb{E}_1[\pi_2]$, and q_1 into (1). Using the period-2 equilibrium:

$$\mathbb{E}_1[y_2] = 0, \quad (27)$$

$$\mathbb{E}_1[\pi_2] = \frac{\kappa\varepsilon_c}{1 + \lambda\kappa^2} \quad (\text{FG}), \quad \mathbb{E}_1[\pi_2] = \kappa \mathbb{E}\left[\frac{\varepsilon_c}{1 + \lambda\kappa^2}\right] \quad (\text{OP}), \quad (28)$$

where in (27) we used $\mathbb{E}[y_2^*] = 0$ under both regimes (since shocks are mean-zero from the period-1 perspective, using $\mathbb{E}[\varepsilon_d] = 0$ and the structure of y_2^*).

Substituting into the period-1 IS curve:

$$y_1^r = -\sigma(i_1 - \mathbb{E}_1^r[\pi_2]) - \alpha(i_1 + \mathbb{E}_1^r[i_2]) = -\tilde{\sigma} i_1 + \sigma \mathbb{E}_1^r[\pi_2] - \alpha \mathbb{E}_1^r[i_2]. \quad (29)$$

Period-1 NKPC.

$$\pi_1^r = \beta \mathbb{E}_1^r[\pi_2] + \kappa y_1^r. \quad (30)$$

Equations (29)–(30) define the period-1 allocation under each regime as a function of i_1 and the regime-specific expectation $\mathbb{E}_1^r[i_2]$. The central bank chooses i_1 to minimise expected total loss, taking (15) as the period-2 continuation.

3 Welfare Comparison

We compare the *expected* total loss $\mathbb{E}[\mathcal{L}]$ under forward guidance and opacity, where expectations are taken over the realisations of both shocks $(\varepsilon_d, \varepsilon_c)$, and—in the opacity case—over the CB’s type.

3.1 Period-2 Losses

The period-2 loss for a central bank of type λ under either regime is $\mathcal{L}_2(\lambda) = (y_2^*)^2 + \lambda(\pi_2^*)^2$. Substituting (21) and (22):

$$\mathcal{L}_2(\lambda) = \frac{(1 + \lambda^2 \kappa^2) \varepsilon_{d,2}^2 - 2\lambda \kappa \varepsilon_{d,2} \varepsilon_c + \lambda(1 + \kappa^2) \varepsilon_c^2}{(1 + \lambda \kappa^2)^2}. \quad (31)$$

Taking expectations over $(\varepsilon_d, \varepsilon_c)$ with $\mathbb{E}[\varepsilon_d] = \mathbb{E}[\varepsilon_c] = 0$, $\text{Var}(\varepsilon_d) = \sigma_d^2$, $\text{Var}(\varepsilon_c) = \sigma_c^2$, and $\text{Cov}(\varepsilon_d, \varepsilon_c) = 0$:

$$\mathbb{E}[\mathcal{L}_2(\lambda)] = \frac{(1 + \lambda^2 \kappa^2) \sigma_d^2 + \lambda(1 + \kappa^2) \sigma_c^2}{(1 + \lambda \kappa^2)^2}. \quad (32)$$

3.2 Exchange Rate Misalignment and Period-1 Losses

Under forward guidance, private agents know the central bank’s type and form correct expectations of the period-2 rate. The period-1 exchange rate is:

$$q_1^{\text{FG}}(\lambda) = -i_1 - \phi_c(\lambda) \varepsilon_c. \quad (33)$$

Under opacity, agents use the prior-weighted average:

$$q_1^{\text{OP}} = -i_1 - \bar{\phi}_c \varepsilon_c. \quad (34)$$

Define the *exchange rate misalignment* under opacity as:

$$\Delta q_1(\lambda) \equiv q_1^{\text{FG}}(\lambda) - q_1^{\text{OP}} = -(\phi_c(\lambda) - \bar{\phi}_c) \varepsilon_c. \quad (35)$$

Since $\phi_c(\lambda)$ is strictly increasing, $\Delta q_1(\lambda_H) < 0$ and $\Delta q_1(\lambda_L) > 0$: the exchange rate is too weak (depreciated) under opacity when the true type is hawkish and too strong (appreciated) when the true type is dovish.

This misalignment feeds into the period-1 IS curve and NKPC (equations (29)–(30)), generating a period-1 welfare loss under opacity relative to forward guidance.

3.3 Main Result

Proposition 2 (Forward Guidance Weakly Dominates Opacity). *Let $\mathcal{L}^r$ denote total expected loss under regime $r \in \{\text{FG}, \text{OP}\}$, where the expectation is taken over shocks and, under opacity, over the prior distribution of types. Then:*

$$\mathbb{E}[\mathcal{L}^{\text{FG}}] \leq \mathbb{E}[\mathcal{L}^{\text{OP}}], \quad (36)$$

with strict inequality whenever $p \in (0, 1)$ and $\lambda_H \neq \lambda_L$ and $\sigma_c^2 > 0$.

Proof sketch. The total expected loss under each regime decomposes as:

$$\mathbb{E}[\mathcal{L}^r] = \mathbb{E}[\mathcal{L}_1^r] + \beta \mathbb{E}[\mathcal{L}_2^r]. \quad (37)$$

Period-2 comparison. Under FG, the period-2 policy is the same function of shocks as under OP (the bank re-optimises in period 2 regardless of the period-1 announcement). However,

under FG the central bank's known type is λ , so:

$$\mathbb{E}[\mathcal{L}_2^{\text{FG}}] = \mathbb{E}[\mathcal{L}_2(\lambda)]. \quad (38)$$

Under OP, the ex-ante expected period-2 loss integrates over the type distribution:

$$\mathbb{E}[\mathcal{L}_2^{\text{OP}}] = p \mathbb{E}[\mathcal{L}_2(\lambda_H)] + (1 - p) \mathbb{E}[\mathcal{L}_2(\lambda_L)]. \quad (39)$$

Both expressions equal the same weighted average *ex ante*—period-2 losses are identical in expectation across regimes. The difference arises solely from the period-1 losses.

Period-1 comparison. From (29) and (30), the period-1 output gap and inflation under regime r are functions of $\mathbb{E}_1^r[i_2]$. Under FG, $\mathbb{E}_1^{\text{FG}}[i_2] = \phi_c(\lambda)\varepsilon_c$ (correct). Under OP, $\mathbb{E}_1^{\text{OP}}[i_2] = \bar{\phi}_c\varepsilon_c$ (an average that may diverge from the true value). For a given i_1 , the period-1 loss under opacity exceeds that under FG by:

$$\mathcal{L}_1^{\text{OP}} - \mathcal{L}_1^{\text{FG}} \propto (\phi_c(\lambda) - \bar{\phi}_c)^2 \varepsilon_c^2. \quad (40)$$

Taking expectations over ε_c and the prior over λ :

$$\mathbb{E}[\mathcal{L}_1^{\text{OP}}] - \mathbb{E}[\mathcal{L}_1^{\text{FG}}] \propto \text{Var}_\lambda[\phi_c(\lambda)] \cdot \sigma_c^2 \geq 0, \quad (41)$$

where $\text{Var}_\lambda[\phi_c(\lambda)] = p(1 - p)(\phi_c(\lambda_H) - \phi_c(\lambda_L))^2 > 0$ whenever $p \in (0, 1)$ and $\lambda_H \neq \lambda_L$. This is strictly positive when $\sigma_c^2 > 0$, completing the argument. $\square$

Economic interpretation. The welfare gain from forward guidance has two additive components:

- (i) **Exchange rate channel.** By revealing its type, the central bank allows markets to correctly price the path of the real exchange rate from period 1 onwards. Under opacity, the exchange rate is anchored to the prior-weighted average response coefficient

$\bar{\phi}_c$, generating a misalignment $\Delta q_1(\lambda)$ that distorts period-1 allocations. This channel is specific to the open-economy setting.

- (ii) **Expectation coordination channel.** Correct knowledge of $\phi_c(\lambda)$ also reduces the variance of expected future inflation $\mathbb{E}_1[\pi_2]$ and future output $\mathbb{E}_1[y_2]$, tightening the period-1 IS curve and NKPC around their optimal values.

Both channels are activated only when cost-push shocks are present ($\sigma_c^2 > 0$) and genuine type uncertainty exists ($p \in (0, 1)$, $\lambda_H \neq \lambda_L$). In the absence of cost-push shocks, demand shocks are accommodated identically by all types, ϕ_c plays no role, and the two regimes are welfare-equivalent.

3.4 The Exchange Rate Amplification Effect

A key prediction is that the welfare advantage of forward guidance is *larger* in more open economies. To see this, note that the misalignment term in (40) involves $(\phi_c(\lambda) - \bar{\phi}_c)^2$, where $\phi_c(\lambda) = \lambda\kappa/[\tilde{\sigma}(1 + \lambda\kappa^2)]$.

Corollary 2.1 (Openness and the Value of Transparency). *The welfare gain from forward guidance is decreasing in the degree of openness α (holding σ fixed).*

Proof. Since $\tilde{\sigma} = \sigma + \alpha$, increasing α raises $\tilde{\sigma}$, which reduces $\phi_c(\lambda)$ for all λ . As $\tilde{\sigma} \rightarrow \infty$, all $\phi_c(\lambda) \rightarrow 0$ and the welfare gap (41) vanishes. The amplification is therefore *non-monotone*: openness amplifies the transmission of the policy rate to output (via $\tilde{\sigma}$), but it also compresses the dispersion of reaction functions across types. □ □

Remark. Corollary 2.1 has a substantive implication: very open economies may have less to gain from formal forward guidance, because the exchange rate channel already transmits the policy stance efficiently—the market can “read” the central bank’s intentions from the current exchange rate. This is consistent with the observation that several highly open economies (e.g. Hong Kong, Singapore) operate under monetary frameworks that rely less on explicit rate-path guidance.

4 Extensions

4.1 Intermediate Communication Regimes

The binary comparison between forward guidance and opacity abstracts from a range of communication practices used by actual central banks. Two intermediate regimes are of particular interest. The *Swiss National Bank* (SNB) publishes projections of inflation and output conditional on the policy rate remaining constant throughout the forecast horizon. The *Bank of England* (BOE) publishes projections conditional on the market-implied path of future interest rates—the path inferred from financial market instruments at the time of the announcement. In both cases, the central bank publishes rich economic forecasts, but it does not directly announce its intended rate path.

The SNB regime. Let $\bar{i}$ be the reference rate at which the SNB conditions its forecast; in practice this is the current period-1 rate i_1 .² The announced period-2 forecasts are:

$$\hat{y}_2^{\text{SNB}} \equiv \mathbb{E}_1[y_2 \mid i_2 = i_1] = -\tilde{\sigma} i_1, \quad (42)$$

$$\hat{\pi}_2^{\text{SNB}} \equiv \mathbb{E}_1[\pi_2 \mid i_2 = i_1] = -\kappa \tilde{\sigma} i_1 + \varepsilon_c, \quad (43)$$

where we have used $\mathbb{E}_1[\varepsilon_{d,2}] = 0$ and the reduced-form expressions (10)–(7).

The BOE regime. Under opacity, the private sector expects $i_2 = \bar{\phi}_c \varepsilon_c$ (equation (24)). The BOE conditions its forecast on this market-implied path:

$$\hat{y}_2^{\text{BOE}} \equiv \mathbb{E}_1[y_2 \mid i_2 = \bar{\phi}_c \varepsilon_c] = -\tilde{\sigma} \bar{\phi}_c \varepsilon_c, \quad (44)$$

$$\hat{\pi}_2^{\text{BOE}} \equiv \mathbb{E}_1[\pi_2 \mid i_2 = \bar{\phi}_c \varepsilon_c] = (1 - \kappa \tilde{\sigma} \bar{\phi}_c) \varepsilon_c. \quad (45)$$

²The SNB’s conditioning assumption has evolved over time, but the essential feature is that the rate is held fixed at a known, publicly observed level—not the bank’s intended future path.

The conditioning assumption is endogenous to market expectations: private agents form $\mathbb{E}_1[i_2]$, the BOE conditions on this, and equilibrium requires consistency between the two. Below we verify that the opacity expectation is the unique self-consistent fixed point.

Proposition 3 (Intermediate Regimes Are Welfare-Equivalent to Opacity). *Under either the SNB or the BOE communication regime, the private sector's equilibrium expectation of the period-2 rate satisfies $\mathbb{E}_1[i_2] = \bar{\phi}_c \varepsilon_c$, the period-1 allocation is identical to that under opacity, and welfare satisfies*

$$\mathcal{L}^{\text{SNB}} = \mathcal{L}^{\text{BOE}} = \mathcal{L}^{\text{OP}}. \quad (46)$$

Proof. SNB regime. Inspect (42)–(43). Both announced forecasts depend only on i_1 , $\tilde{\sigma}$, κ , and ε_c . Of these, $\tilde{\sigma}$ and κ are common knowledge, i_1 is publicly observed in period 1, and ε_c is publicly observed by assumption (Section 2.3). The announcements are therefore redundant given the agents' information set: they contain no information about λ beyond what is already known. Agents continue to use the prior to form $\mathbb{E}_1^{\text{SNB}}[i_2] = \bar{\phi}_c \varepsilon_c$, as under opacity.

BOE regime. Suppose agents form expectation $\mathbb{E}_1[i_2] = m \varepsilon_c$ for some coefficient m to be determined. The BOE conditions on $i_2 = m \varepsilon_c$ and announces (44)–(45) with $\bar{\phi}_c$ replaced by m . These announcements depend only on m , $\tilde{\sigma}$, κ , and ε_c —none of which reveal λ . Agents learn nothing about λ and update m only through prior beliefs: $m = \bar{\phi}_c$. The fixed point $m = \bar{\phi}_c$ is therefore unique, confirming $\mathbb{E}_1^{\text{BOE}}[i_2] = \bar{\phi}_c \varepsilon_c$.

Since both intermediate regimes induce the same period-1 expectation as opacity, the period-1 exchange rate, IS curve, and NKPC (equations (25)–(30)) are identical. Period-2 policy is unaffected by period-1 announcements. Welfare is therefore identical to opacity.

□

□

The key mechanism. The result rests on a single structural observation: the central bank's private information λ governs its *reaction function*—how much it raises the rate in response to cost-push shocks—not the structural equations that describe how the econ-

omy responds to a given rate. The IS and Phillips curves (with slopes $\tilde{\sigma}$ and κ) are common knowledge. Conditional forecasts, by construction, fix the rate path and apply these common-knowledge structural equations; they therefore mechanically strip out the type-dependent information. Only direct communication about the intended rate path encodes λ .

Welfare hierarchy. Combining Propositions 2 and 3:

$$\mathcal{L}^{\text{FG}} \leq \mathcal{L}^{\text{SNB}} = \mathcal{L}^{\text{BOE}} = \mathcal{L}^{\text{OP}}, \quad (47)$$

with strict inequality on the left whenever $p \in (0, 1)$, $\lambda_H \neq \lambda_L$, and $\sigma_c^2 > 0$. The two intermediate regimes, while reflecting genuine and substantive differences in central bank practice, are welfare-equivalent to full opacity in this model. The implication is stark: what matters for welfare is not the *volume* of information published, but whether the publication encodes information about the central bank’s *intended rate path* and, through it, its type.

Remark: private cost-push shocks. The equivalence result depends on ε_c being publicly observed. If, instead, ε_c were private information of the central bank, SNB-style and BOE-style announcements would reveal ε_c to private agents (via (43) and (45)), yielding a strict welfare gain relative to full opacity. In this richer setting, the intermediate regimes would occupy a genuinely intermediate position in the welfare hierarchy, transmitting information about the central bank’s macroeconomic assessment without disclosing its reaction function. We treat this as a natural extension.

4.2 Welfare-Improving Scenario Communication

Proposition 3 identifies a sharp boundary: conditional economic forecasts are welfare-neutral relative to opacity, while explicit rate-path announcements strictly improve welfare. Between these extremes lies *scenario communication*—publishing the intended rate path under mul-

tuple contingencies, imperfectly or with deliberate ambiguity. We now parametrise these intermediate regimes and characterise the welfare gain.

Definition. A *scenario* is a pair $(\varepsilon_c^{(k)}, \hat{i}_2^{(k)})$ specifying the central bank's intended period-2 rate conditional on the cost-push shock taking value $\varepsilon_c^{(k)}$. A *scenario set* is a menu of K such pairs $\mathcal{S} = \{(\varepsilon_c^{(k)}, \hat{i}_2^{(k)})\}_{k=1}^K$. The SNB and BOE regimes are special cases with $\hat{i}_2^{(k)}$ fixed independently of λ ; a fully contingent scenario has $\hat{i}_2^{(k)} = \phi_c(\lambda)\varepsilon_c^{(k)}$.

Scenarios and type revelation. Since $i_2^*(\lambda) = \phi_d\varepsilon_d + \phi_c(\lambda)\varepsilon_c$, a scenario $(\varepsilon_c^{(k)}, \hat{i}_2^{(k)})$ with $\hat{i}_2^{(k)} = \phi_c(\lambda)\varepsilon_c^{(k)}$ at any $\varepsilon_c^{(k)} \neq 0$ fully reveals λ (since ϕ_c is strictly monotone), and is therefore informationally equivalent to full forward guidance. Non-contingent scenarios (SNB, BOE) fix $\hat{i}_2^{(k)}$ independently of λ and hence convey zero type information.

The θ -parametrisation. To bridge these extremes we characterise the informativeness of an arbitrary scenario set $\mathcal{S}$ by the *Blackwell informativeness ratio*:

$$\theta(\mathcal{S}) \equiv \frac{\text{Var}_a[\mathbb{E}[\phi_c(\lambda) \mid a]]}{\text{Var}_\lambda[\phi_c(\lambda)]}, \quad (48)$$

where a denotes the signal (announcement) received by private agents and expectations are taken with respect to the equilibrium signal distribution. By the law of total variance, $\theta(\mathcal{S}) \in [0, 1]$:

- $\theta = 0$: no type information is transmitted (opacity, SNB, BOE);
- $\theta = 1$: type is fully revealed (full forward guidance);
- $\theta \in (0, 1)$: partial transmission.

Proposition 4 (Welfare Gain Is Linear in Informativeness). *For any scenario set $\mathcal{S}$ with Blackwell informativeness $\theta(\mathcal{S})$:*

$$\mathcal{L}^{\text{OP}} - \mathcal{L}^{\mathcal{S}} = \theta(\mathcal{S}) (\mathcal{L}^{\text{OP}} - \mathcal{L}^{\text{FG}}). \quad (49)$$

The welfare gain from scenario communication is exactly $\theta(\mathcal{S})$ times the maximum (full forward guidance) gain.

Proof. From the period-1 welfare decomposition (41), the expected loss under any communication regime r satisfies

$$\mathbb{E}[\mathcal{L}_1^r] = \mathbb{E}[\mathcal{L}_1^{\text{FG}}] + c \cdot \mathbb{E}_a[\text{Var}[\phi_c(\lambda) \mid a]] \cdot \sigma_c^2, \quad (50)$$

for some constant $c > 0$ that depends only on the structural parameters. (Under FG the posterior variance is zero; under opacity it is $\text{Var}_\lambda[\phi_c(\lambda)]$.) By the law of total variance:

$$\mathbb{E}_a[\text{Var}[\phi_c(\lambda) \mid a]] = \text{Var}_\lambda[\phi_c(\lambda)] - \text{Var}_a[\mathbb{E}[\phi_c(\lambda) \mid a]] = (1 - \theta) \text{Var}_\lambda[\phi_c(\lambda)]. \quad (51)$$

Substituting (51) into (50):

$$\mathbb{E}[\mathcal{L}_1^S] = \mathbb{E}[\mathcal{L}_1^{\text{FG}}] + c(1 - \theta) \text{Var}_\lambda[\phi_c(\lambda)] \sigma_c^2. \quad (52)$$

At $\theta = 0$ (opacity): $\mathbb{E}[\mathcal{L}_1^{\text{OP}}] = \mathbb{E}[\mathcal{L}_1^{\text{FG}}] + c \text{Var}_\lambda[\phi_c(\lambda)] \sigma_c^2$. Subtracting:

$$\mathbb{E}[\mathcal{L}_1^{\text{OP}}] - \mathbb{E}[\mathcal{L}_1^S] = \theta c \text{Var}_\lambda[\phi_c(\lambda)] \sigma_c^2 = \theta (\mathbb{E}[\mathcal{L}_1^{\text{OP}}] - \mathbb{E}[\mathcal{L}_1^{\text{FG}}]).$$

Period-2 losses are identical across regimes (the central bank re-optimises in period 2 regardless of what it announced in period 1), so the equality carries through to total expected losses. □ □

Canonical implementation: the θ -disclosure rule. A convenient mechanism that achieves any target $\theta \in [0, 1]$ is the θ -disclosure rule: the central bank commits to revealing its true type (announcing $\phi_c(\lambda)\varepsilon_c$) with probability θ , and making no type-revealing statement (announcing $\bar{\phi}_c\varepsilon_c$) with probability $1 - \theta$. In the binary type model, where $\phi_c(\lambda_H)\varepsilon_c \neq \phi_c(\lambda_L)\varepsilon_c$ whenever $\varepsilon_c \neq 0$, agents perfectly infer the regime on the equilibrium

path (a revealing announcement is distinguishable from the prior-mean announcement), so the rule is credibly incentive-compatible.

Under this rule, the economy is in the FG regime with probability θ and in the opacity regime with probability $1 - \theta$. Expected total loss:

$$\mathcal{L}^\theta = \theta \mathcal{L}^{\text{FG}} + (1 - \theta) \mathcal{L}^{\text{OP}}, \quad (53)$$

which immediately yields (49). Equation (53) makes clear that the welfare hierarchy is a line segment connecting the opacity and FG outcomes as θ moves from 0 to 1.

Diminishing returns to signal precision. While the welfare gain is linear in informativeness θ , it exhibits *diminishing returns to signal precision* when the scenario is implemented via a noisy rate announcement. Suppose the central bank announces $s = \phi_c(\lambda)\varepsilon_c + \xi$ where ξ is mean-zero noise with variance $1/\tau$ (precision $\tau \geq 0$). With a Gaussian prior $\phi_c(\lambda) \sim \mathcal{N}(\bar{\phi}_c, \sigma_\phi^2)$, the posterior variance is $\sigma_\phi^2/(1 + \tau\sigma_\phi^2\varepsilon_c^2)$, and the resulting informativeness is:

$$\theta(\tau) = \frac{\tau\sigma_\phi^2\varepsilon_c^2}{1 + \tau\sigma_\phi^2\varepsilon_c^2}, \quad (54)$$

which is increasing and strictly concave in τ . Combining with Proposition 4, the welfare gain as a function of precision is:

$$\mathcal{L}^{\text{OP}} - \mathcal{L}^\tau = \frac{\tau\sigma_\phi^2\varepsilon_c^2}{1 + \tau\sigma_\phi^2\varepsilon_c^2} \cdot (\mathcal{L}^{\text{OP}} - \mathcal{L}^{\text{FG}}). \quad (55)$$

Marginal gains from increasing precision τ are positive throughout but converging to zero as $\tau \rightarrow \infty$. The full FG gain is recovered only in the limit $\tau \rightarrow \infty$. This concavity implies that *once a central bank has committed to some scenario communication, the incremental value of further precision—tighter rate ranges, more scenarios—declines.*

Taxonomy. Proposition 4 implies the following welfare-ordered taxonomy:

1. **Forward guidance** ($\theta = 1$): maximum gain; full type revelation via the rate-path announcement.
2. **Contingent scenarios** ($\theta \in (0, 1)$): intermediate gain = $\theta \times$ FG gain; welfare improvement is proportional to Blackwell informativeness.
3. **Market-conditioned or constant-rate forecasts** ($\theta = 0$): zero gain; non-contingent announcements cannot encode the reaction function.
4. **Opacity** ($\theta = 0$): benchmark; same welfare as SNB/BOE.

Optimal scenario design (absent commitment frictions) requires $\theta = 1$: Proposition 4 shows no interior maximum. If the central bank faces a cost $C(\theta)$ of transparency (legal constraints, political exposure, or equilibrium credibility bounds), the optimal θ^* solves:

$$\theta^* = \arg \max_{\theta \in [0,1]} \theta (\mathcal{L}^{\text{OP}} - \mathcal{L}^{\text{FG}}) - C(\theta), \quad (56)$$

yielding an interior solution $\theta^* \in (0, 1)$ whenever $C'(0) < \mathcal{L}^{\text{OP}} - \mathcal{L}^{\text{FG}}$ and C is convex. In this case the marginal value of transparency equals the marginal cost of disclosure.

4.3 Continuous Type Distribution

The binary type assumption ($\lambda \in \{\lambda_L, \lambda_H\}$) can be relaxed to a continuous distribution $\lambda \sim F$ on $[\lambda_L, \lambda_H]$. Under forward guidance, the announcement $\hat{i}_2(\lambda) = \phi_c(\lambda)\varepsilon_c$ fully reveals λ for any realisation of $\varepsilon_c \neq 0$, since ϕ_c is strictly monotone. The welfare comparison in Proposition 2 extends directly: the welfare gap between regimes equals $\text{Var}_\lambda[\phi_c(\lambda)] \cdot \sigma_c^2$, now computed over the continuous distribution. Richer distributional assumptions (e.g. fat tails for the hawkish type) can affect the quantitative magnitude of the gain.

4.4 Strategic Announcements

In the baseline, the announcement is truthful by assumption. An important extension considers the case in which the central bank is *strategic*: it chooses what to announce to influence period-1 private-sector behaviour, subject to the constraint that the announcement must be credible in a Bayesian equilibrium. Under full commitment, the truthful announcement remains an equilibrium (a separating equilibrium in the signalling game), since the announcement perfectly identifies the type. Under partial commitment, pooling equilibria may arise in which the central bank shades its announcement toward the prior mean, reducing the informational content of forward guidance.

4.5 Risk Premia and UIP Deviations

The baseline model assumes that UIP holds exactly. A standard extension introduces a time-varying risk premium ρ_t in the UIP condition:

$$q_t = \mathbb{E}_t[q_{t+1}] - (i_t - i^*) + \rho_t. \quad (57)$$

If ρ_t is correlated with the central bank's type (e.g. uncertainty about the reaction function generates a term premium), then forward guidance has an additional channel: by resolving type uncertainty, it reduces the risk premium, appreciates the exchange rate, and further improves period-1 allocations. This channel is quantitatively relevant for economies with high exchange rate pass-through to import prices.

5 Conclusion

This paper develops a two-period small open economy New Keynesian model to compare explicit forward guidance with opacity when the central bank's inflation aversion is private information. The main findings are:

- (i) Forward guidance weakly dominates opacity in welfare terms. The gain is strictly positive whenever there is genuine type uncertainty ($p \in (0, 1)$, $\lambda_H \neq \lambda_L$) and cost-push shocks are present ($\sigma_c^2 > 0$).
- (ii) The welfare gain operates through two channels: an exchange rate channel—specific to the open-economy setting—and an expectation coordination channel. Both channels reduce the misalignment between private-sector beliefs and the central bank’s actual reaction function.
- (iii) The value of transparency is non-monotone in the degree of openness: greater openness amplifies policy transmission but also compresses the cross-type dispersion in reaction functions, potentially reducing the informational value of the announcement.

[Conclusion will be extended once quantitative results are available.]